Runbook — clean-handoff voice agent (Brightside Dental demo)

> **System:** Vapi web-based voice assistant + Node/TS Express backend (mock CRM, webhook handlers, dashboard)
> **Owner:** Guillermo Mustafa
> **Applies to:** production deployment (Render) and local development (ngrok)

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1. Overview

Architecture, in call order:

1. Caller opens `/demo` and starts a browser mic call (Vapi Web SDK — stands in for a PSTN number; a production deployment would attach a Twilio number to the same assistant with no backend changes).
2. Vapi runs the assistant (STT → LLM → TTS) using the system prompt in `assistant/system-prompt.md`.
3. Mid-call, the assistant calls custom tools → `POST /vapi/tool-calls` on this backend (auth: server secret header). Tools: `lookupPatient`, `checkAvailability`, `bookAppointment`, `createLead`, `escalateToHuman`.
4. On escalation, `escalateToHuman` writes a structured `Handoff` record; `/dashboard` renders it for the human agent.
5. After every call, Vapi fires `end-of-call-report` → `POST /vapi/events` → `CallLog` record.

Data store: `server/data/db.json` (lowdb). Seeded by `server/src/data/seed.ts` on first boot.

2. Trigger conditions — when to use this runbook

- Demo call connects but the agent can't book / says it "can't check availability" (tool failures)
- Escalations happen in-call but nothing appears on `/dashboard`
- `/demo` page can't start a call at all
- Webhook auth failures (401s in backend logs)
- Backend down or unreachable
- Corrupted or stale data in `db.json`

3. Diagnostic steps

Run in order; stop at the first failing step.

#	Check	How	Healthy result
1	Backend up	`GET https://<backend-url>/health`	`{"ok":true,...}`
2	Backend logs	Render dashboard → service → Logs (or local terminal)	`[tool:*]` and `[event:*]` lines on recent calls
3	Recent handoffs/calls	`GET /handoffs`, `GET /calls`	JSON arrays with recent records
4	Vapi side	Vapi dashboard → Call Logs → open the failing call by `callId`	Tool calls show request + response; errors show HTTP status from our backend
5	Auth	Compare the credential configured on the Vapi tools/server URL vs `VAPI_SERVER_SECRET` env var on the backend	Must match exactly (backend accepts it as `X-Vapi-Secret` or `Authorization: Bearer`)
6	Tool reachability	`curl -X POST <backend-url>/vapi/tool-calls -H "X-Vapi-Secret: <secret>" ...`	`{"results":[{"toolCallId":"t1","result":"..."}]}`

4. Resolution steps

Symptom	Fix
401 in backend logs on tool calls	Secret mismatch — update the Vapi credential or the `VAPI_SERVER_SECRET` env var so they match, then redeploy/restart
Tool calls never arrive at backend	Vapi tools point at a stale URL (old ngrok URL is the classic cause in dev) — update each tool's `server.url` and the assistant's server URL in Vapi
Tools time out (~5s+)	Check Render logs for slow responses; restart the service (Render → Manual Deploy → Restart); tools must respond well under Vapi's timeout
Escalation spoken but no dashboard record	Check backend logs for `[tool:escalateToHuman]` — if the tool rejected the payload (missing `transcriptSummary` / `escalationReason`), the LLM under-filled arguments: tighten/restore the "How to escalate" section of the system prompt
Agent asks the same failed	Prompt drift — restore the "No-input / no-match handling" section of `assistant/system-prompt.md` verbatim (this behavior is prompt-enforced, see §7)

question 3+ times	
`/demo` won't start a call	Check browser console: missing/wrong `VAPI_PUBLIC_KEY` or `ASSISTANT_ID`, or mic permission denied
Backend crash-looping	Render logs → if `db.json` is corrupted, see rollback below

5. Rollback

- **Code:** Render redeploys on push; roll back via Render dashboard → Deploys → "Rollback to this deploy" on the last known-good build (or `git revert` + push).
- **Data:** delete `server/data/db.json` and restart — the server re-seeds from `seed.ts` automatically. Demo data is disposable by design.
- **Assistant config:** the versioned source of truth is `assistant/` in this repo — re-paste `system-prompt.md` and re-check tool schemas against `assistant/tools/*.json` if dashboard edits caused a regression.

6. Escalation matrix

> Illustrative for the demo — in a real engagement, fill with the client's actual contacts/SLAs.

Level	Who	When	Channel	Response SLA
L1	On-call developer (Guillermo)	Any trigger in §2	Direct message	1 business hour
L2	Vapi support	Confirmed platform-side failure (calls not connecting, STT/TTS errors visible in Vapi call logs)	support.vapi.ai	Per Vapi plan
L3	Render support/status	Backend healthy locally but unreachable when deployed	status.render.com → ticket	Per Render plan

In-call escalation (AI → human) is not an incident — it's the designed behavior documented in the call-flow diagram.

7. Known limitations (deliberate, documented)

- **Retry counting is prompt-enforced.** Vapi has no native "max N no-match attempts" primitive like DTMF IVRs; the 2-retry rule lives in the system prompt and is only as reliable as LLM instruction-following. The backend compensates by validating every handoff payload (rejects escalations without a summary/reason), so a malformed escalation fails loudly instead of silently.

- **Handoff field values are LLM-populated.** `transcriptSummary`, `sentiment` and `dataCollectedSoFar` are filled by the model from conversation context — the JSON schema marks the critical ones as required, and the backend re-validates.
- **Web-only demo:** no PSTN number attached (cost decision). The webhook architecture is identical with a Twilio number attached in Vapi; nothing in this backend changes.

8. Communication templates

Handoff notification (to front desk / client team):

> New escalated call — {{urgency}} priority.
> Caller: {{caller.name}} ({{caller.phone}}), {{intent}}.
> Reason: {{escalationReason}}. Summary: "{{transcriptSummary}}"
> Full context: {{dashboard-url}}/handoffs/{{id}} — do not ask the caller to repeat information already listed there.

Callback opener (for the human agent calling back):

> "Hi {{caller.name}}, this is Brightside Dental returning your call — I have your details in front of me: you were calling about {{reasonForVisit}}. Let's pick up right where you left off."

9. Version history

Version	Date	Change
0.1	2026-07-19	Initial build: assistant + backend + dashboard + docs